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Exp9: Implement a search functionality to filter API results dynamically.

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0">

   <title>Dynamic User Explorer | Exp 9</title>

   <meta name="description" content="A dynamic user exploration tool fetching data from a public
API with real-time filtering.">

   <link rel="stylesheet" href="style.css">

   <link rel="preconnect" href="https://fonts.googleapis.com">

   <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

   <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap"
rel="stylesheet">

</head>

<body>

   <div class="container">

      <header>

         <h1>User Explorer</h1>

         <p>Real-time data synchronization with public API endpoints</p>

      </header>


      <main>

         <div class="search-container">

            <input type="text" id="search-input" placeholder="Search users by name or email..."
autocomplete="off">

         </div>


         <div id="loader" class="loader" style="display: none;"></div>

         <div id="user-grid" class="user-grid"></div>
```

```
        </main>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

Style.css:

```
:root {
    --primary: #6366f1;
    --primary-hover: #4f46e5;
    --bg-gradient: linear-gradient(135deg, #0f172a 0%, #1e1b4b 100%);
    --card-bg: rgba(255, 255, 255, 0.05);
    --card-border: rgba(255, 255, 255, 0.1);
    --text-main: #f8fafc;
    --text-muted: #94a3b8;
    --glass-blur: blur(12px);
}

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: 'Inter', system-ui, -apple-system, sans-serif;
}

body {
    background: var(--bg-gradient);
    min-height: 100vh;
    color: var(--text-main);
    padding: 2rem;
```

```
    display: flex;
    flex-direction: column;
    align-items: center;
}

.container {
    max-width: 1000px;
    width: 100%;
}

header {
    text-align: center;
    margin-bottom: 3rem;
    animation: fadeInDown 0.8s ease-out;
}

h1 {
    font-size: 2.5rem;
    font-weight: 800;
    background: linear-gradient(to right, #818cf8, #c084fc);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    margin-bottom: 0.5rem;
}

p {
    color: var(--text-muted);
}

.search-container {
    position: relative;
```

```
    max-width: 600px;
    margin: 0 auto 3rem;
    animation: fadeIn 1s ease-out;
}

#search-input {
    width: 100%;
    padding: 1.25rem 1.5rem;
    background: var(--card-bg);
    border: 1px solid var(--card-border);
    border-radius: 1rem;
    color: white;
    font-size: 1.1rem;
    backdrop-filter: var(--glass-blur);
    transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06);
}

#search-input:focus {
    outline: none;
    border-color: var(--primary);
    box-shadow: 0 0 0 4px rgba(99, 102, 241, 0.2), 0 10px 15px -3px rgba(0, 0, 0, 0.1);
    transform: translateY(-2px);
}

.user-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
    gap: 1.5rem;
    width: 100%;
}
```

```
.user-card {
    background: var(--card-bg);
    border: 1px solid var(--card-border);
    padding: 1.5rem;
    border-radius: 1.25rem;
    backdrop-filter: var(--glass-blur);
    transition: all 0.3s ease;
    animation: cardAppear 0.5s ease-out backwards;
}

.user-card:hover {
    transform: translateY(-5px);
    border-color: rgba(99, 102, 241, 0.5);
    background: rgba(255, 255, 255, 0.08);
}

.user-card h3 {
    margin-bottom: 0.5rem;
    font-size: 1.25rem;
    color: #e2e8f0;
}

.user-card p {
    font-size: 0.9rem;
    line-height: 1.5;
}

.user-card .email {
    color: var(--primary);
    font-weight: 500;
```

```
}

.user-card .company {
    margin-top: 1rem;
    font-size: 0.8rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    color: #94a3b8;
}

/* Animations */
@keyframes fadeInDown {
    from { opacity: 0; transform: translateY(-20px); }
    to { opacity: 1; transform: translateY(0); }
}

@keyframes fadeIn {
    from { opacity: 0; }
    to { opacity: 1; }
}

@keyframes cardAppear {
    from { opacity: 0; transform: scale(0.9); }
    to { opacity: 1; transform: scale(1); }
}

/* Loading Spinner */
.loader {
    width: 48px;
    height: 48px;
    border: 5px solid #FFF;
```

```
    border-bottom-color: var(--primary);
    border-radius: 50%;
    display: inline-block;
    box-sizing: border-box;
    animation: rotation 1s linear infinite;
    margin: 2rem auto;
}

@keyframes rotation {
    0% { transform: rotate(0deg); }
    100% { transform: rotate(360deg); }
}

.no-results {
    text-align: center;
    grid-column: 1 / -1;
    padding: 3rem;
    color: var(--text-muted);
    font-style: italic;
}
```

Script.js:

```
// Global state to store API data
let users = [];
const API_URL = 'https://jsonplaceholder.typicode.com/users';

// DOM Elements
const userGrid = document.getElementById('user-grid');
const searchInput = document.getElementById('search-input');
const loader = document.getElementById('loader');
```

```
/**
 * Fetches data from the public API
 */
async function fetchUsers() {
    try {
        showLoader(true);
        const response = await fetch(API_URL);
        if (!response.ok) throw new Error('Network response was not ok');

        users = await response.json();
        displayUsers(users);
    } catch (error) {
        console.error('Fetch error:', error);
        userGrid.innerHTML = `<p class="no-results">Error fetching data. Please try again later.</p>`;
    } finally {
        showLoader(false);
    }
}

/**
 * Displays user cards in the grid
 * @param {Array} usersToDisplay
 */
function displayUsers(usersToDisplay) {
    userGrid.innerHTML = '';

    if (usersToDisplay.length === 0) {
        userGrid.innerHTML = '<p class="no-results">No users match your search criteria.</p>';
        return;
    }
```

```
    usersToDisplay.forEach((user, index) => {
        const card = document.createElement('div');
        card.className = 'user-card';
        card.style.animationDelay = `${index * 0.05}s`;

        card.innerHTML = `
            <h3>${user.name}</h3>
            <p class="email">@${user.username}</p>
            <p>${user.email}</p>
            <p>${user.address.city}, ${user.address.suite}</p>
            <div class="company">
                <strong>Company:</strong> ${user.company.name}
            </div>
        `;
        userGrid.appendChild(card);
    });
}

/**
 * Filters the users array based on search input
 */
function handleSearch(event) {
    const searchTerm = event.target.value.toLowerCase();

    const filteredUsers = users.filter(user => {
        return (
            user.name.toLowerCase().includes(searchTerm) ||
            user.email.toLowerCase().includes(searchTerm) ||
            user.username.toLowerCase().includes(searchTerm)
        );
    });
```

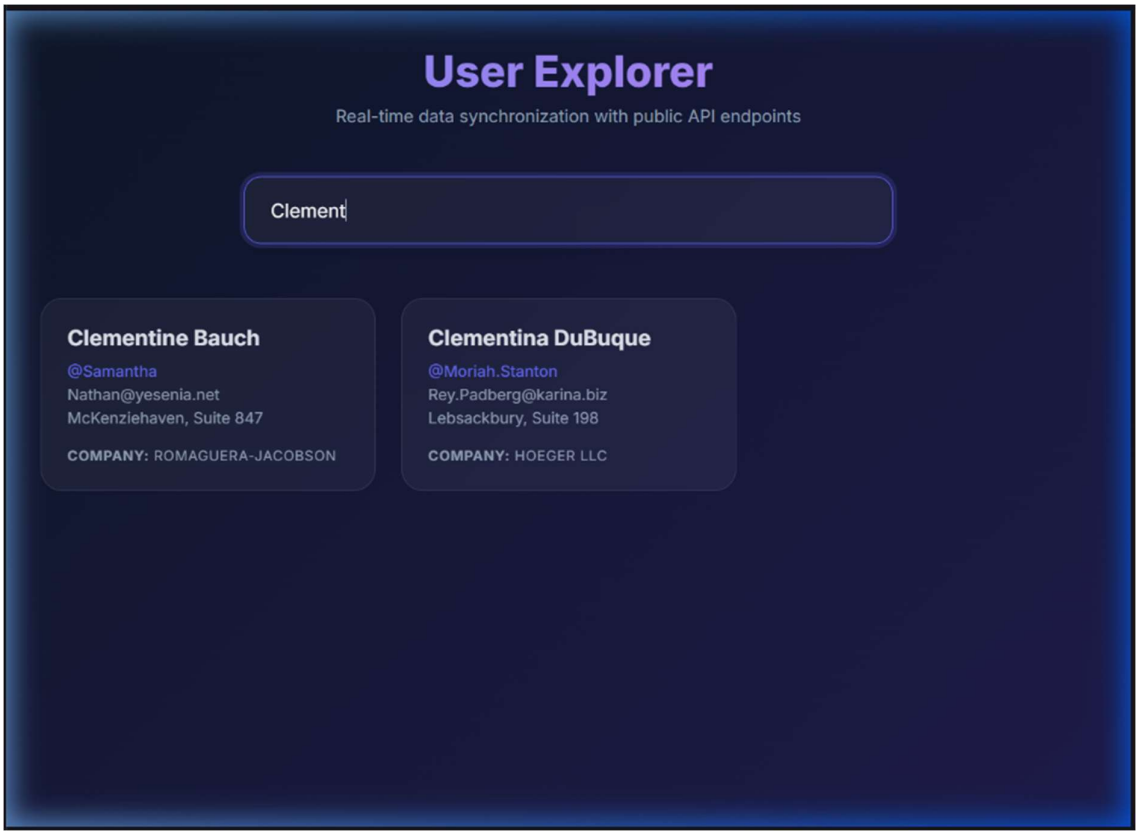
```
    displayUsers(filteredUsers);
}

/**
 * Controls visibility of the loader
 * @param {boolean} isLoading
 */
function showLoader(isLoading) {
    loader.style.display = isLoading ? 'block' : 'none';
}

// Event Listeners
searchInput.addEventListener('input', handleSearch);

// Initialize application
document.addEventListener('DOMContentLoaded', fetchUsers);
```

Output:


User Explorer
Real-time data synchronization with public API endpoints
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